

Appendix A - South-west Corner (Eastern arm) Marine Park Natural Values Report

South-west Marine Parks Network

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Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

Date	Campaign name	Method	Areas	Samples	Habitat	Fish	
					Count	Count	Length
Jan-Nov 2022	2022-11 Investigator stereo-BOSS	BOSS	Coastal waters, NPZ, SPZ	132	131	-	-
Nov 2022	2022-11 Investigator stereo-BRUVs	BRUV	NPZ	38	35	33	30
May 2024	2024-05 Wudjari stereo-BRUVs	BRUV	Coastal waters, NPZ, SPZ	228	228	221	216

Benthic ecosystem extent

Observations

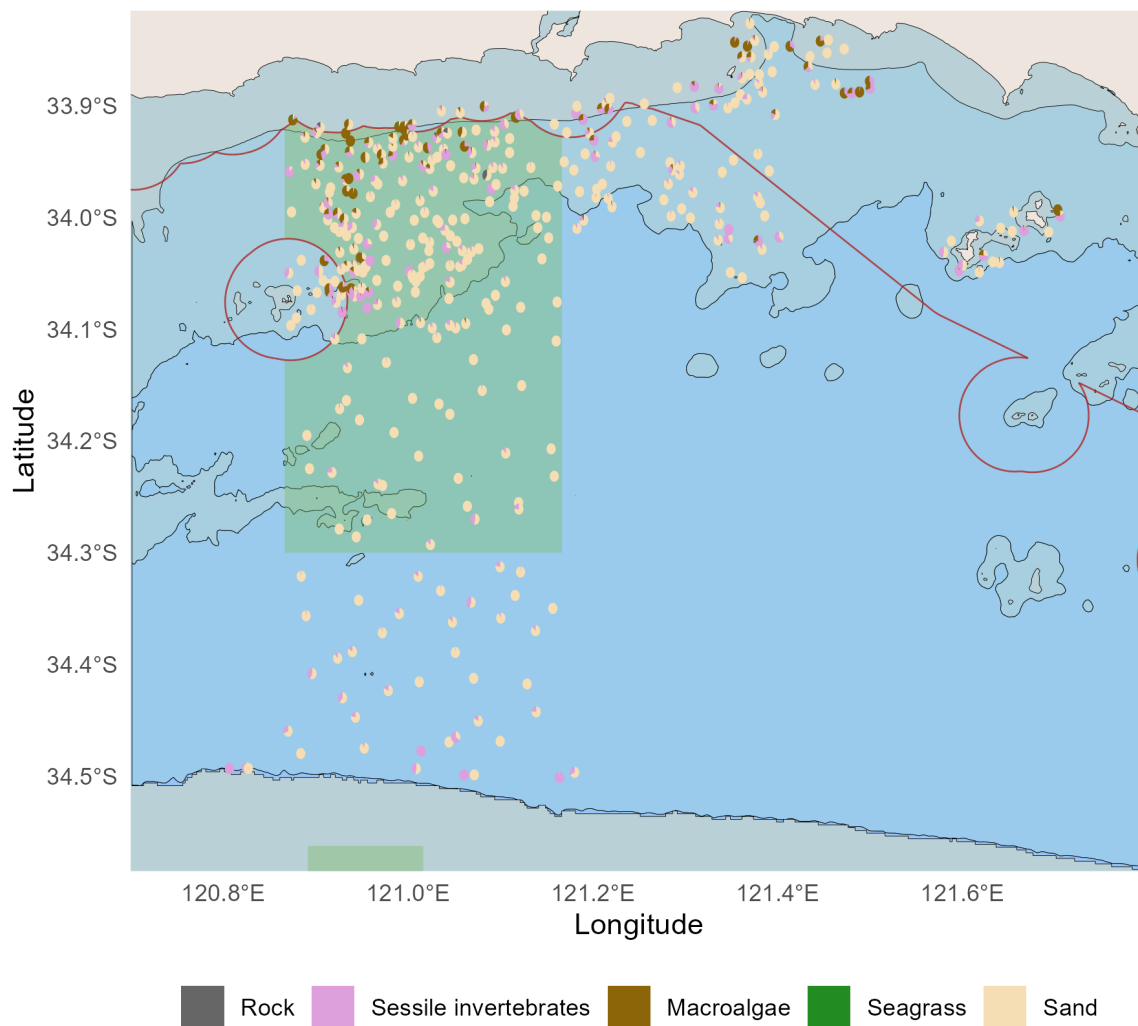


Figure 1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

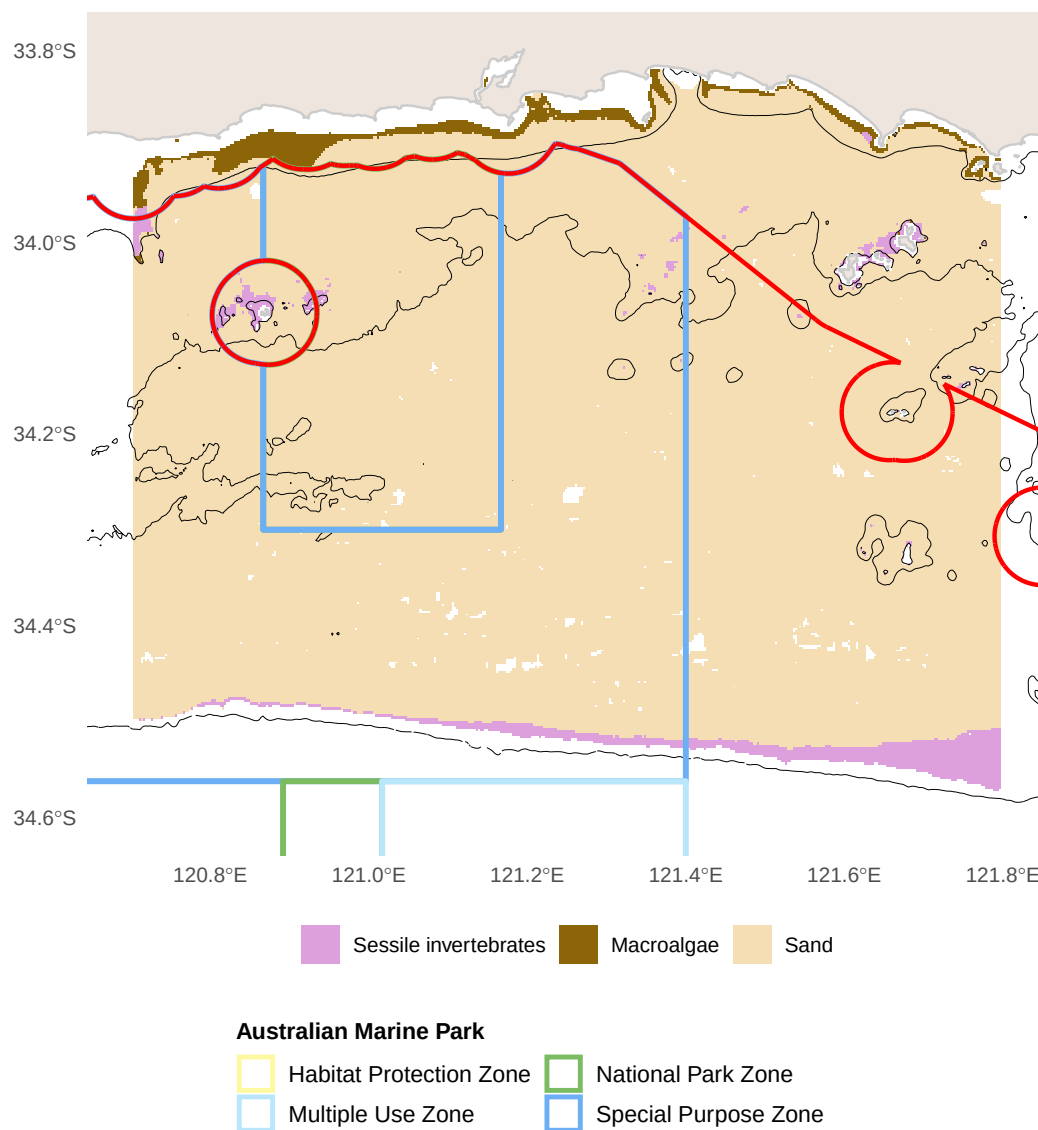


Figure 2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

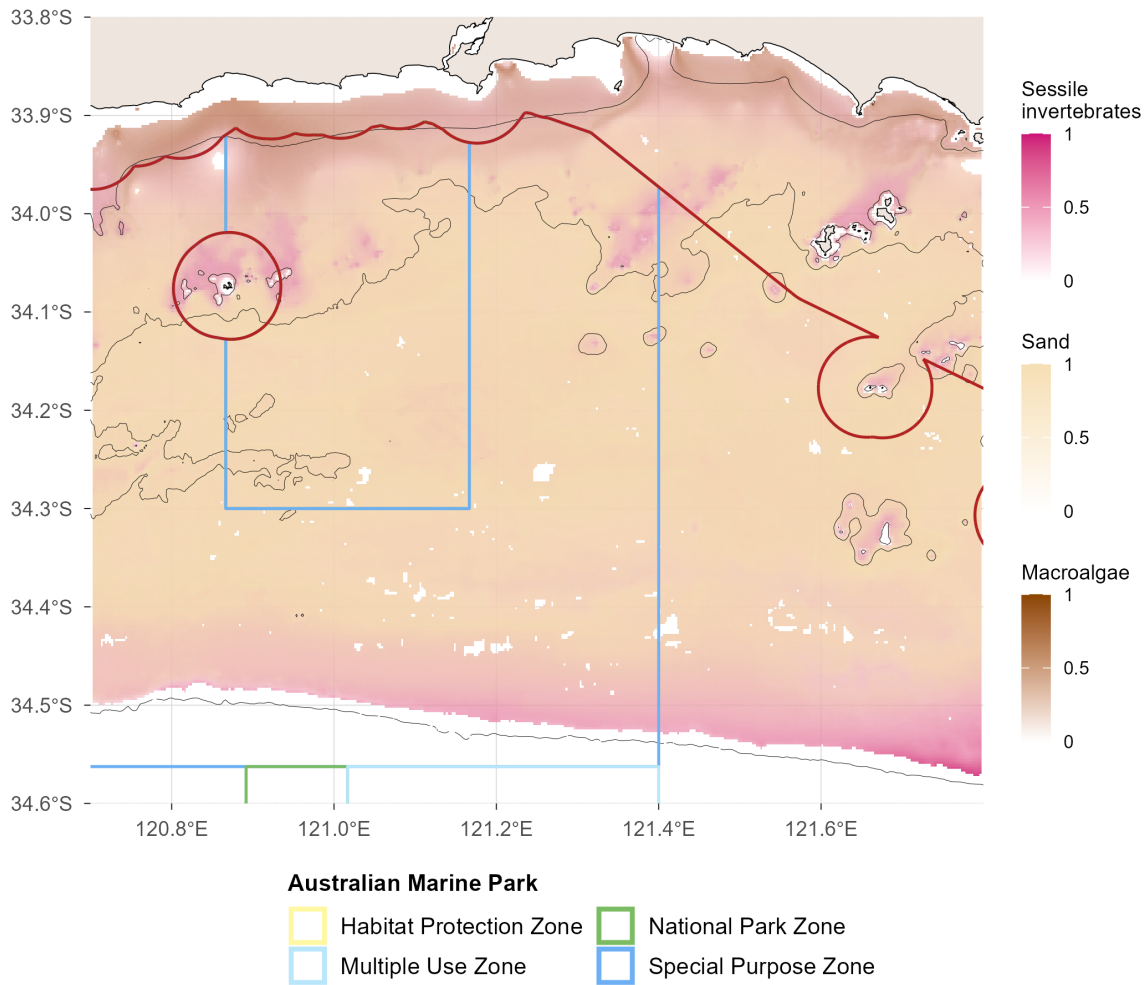


Figure 3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

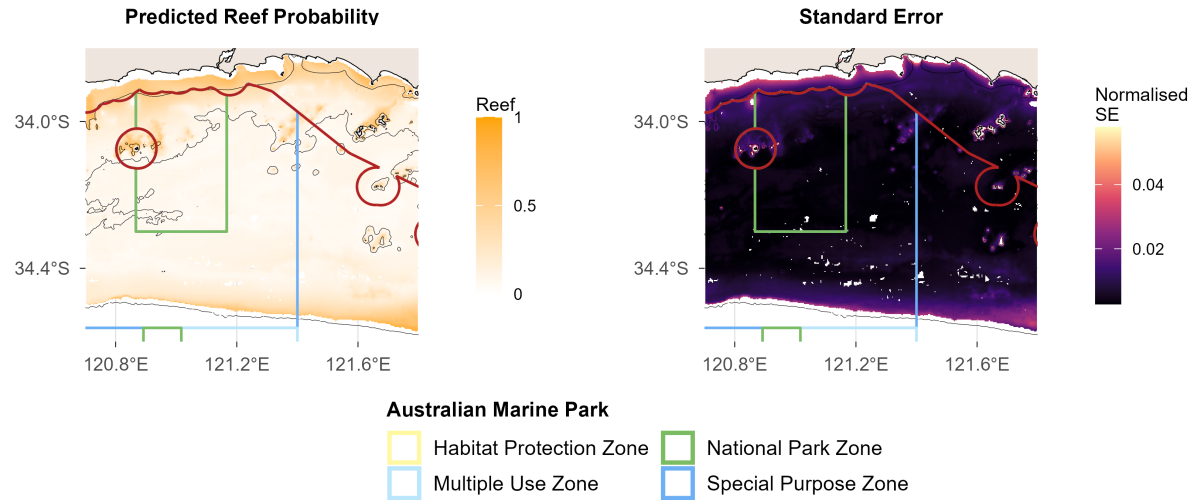


Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Sand

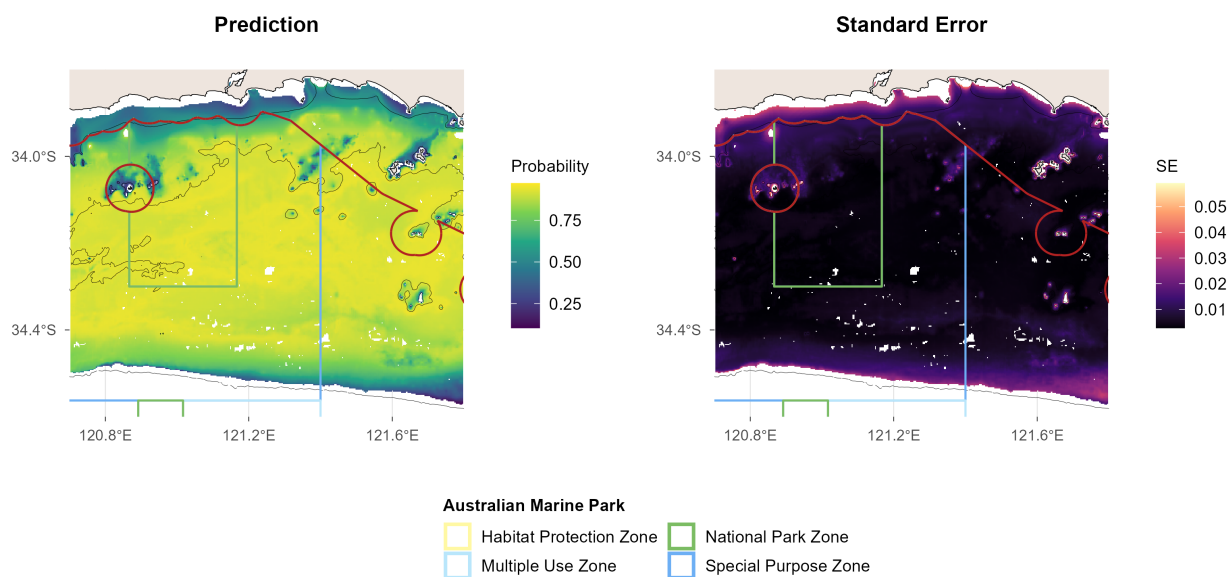


Figure 5: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Macroalgae

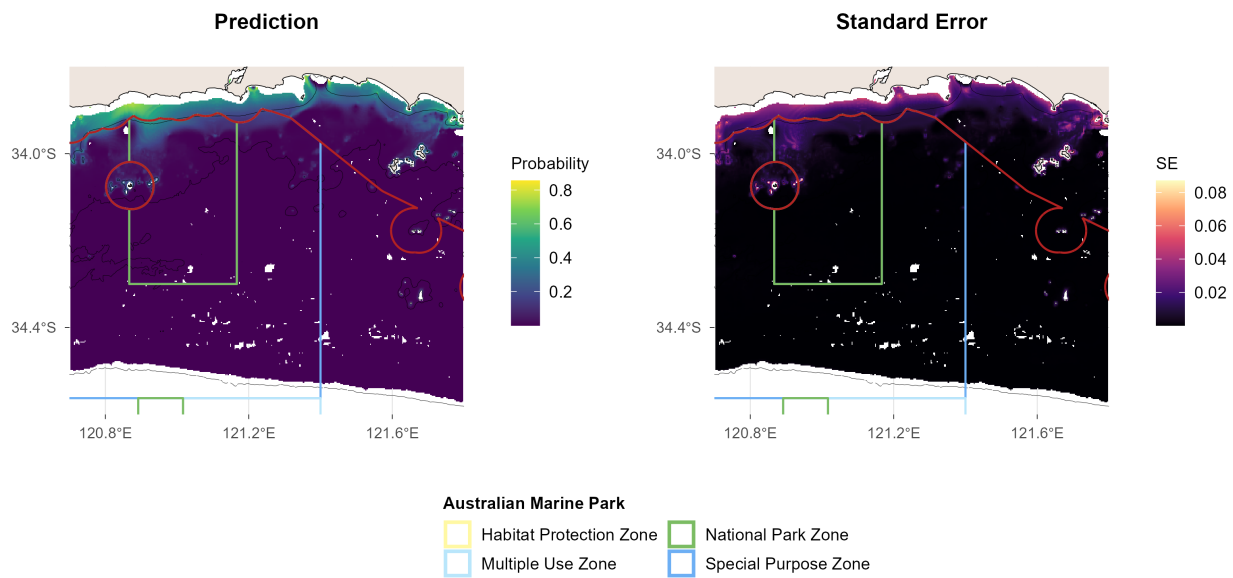


Figure 6: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

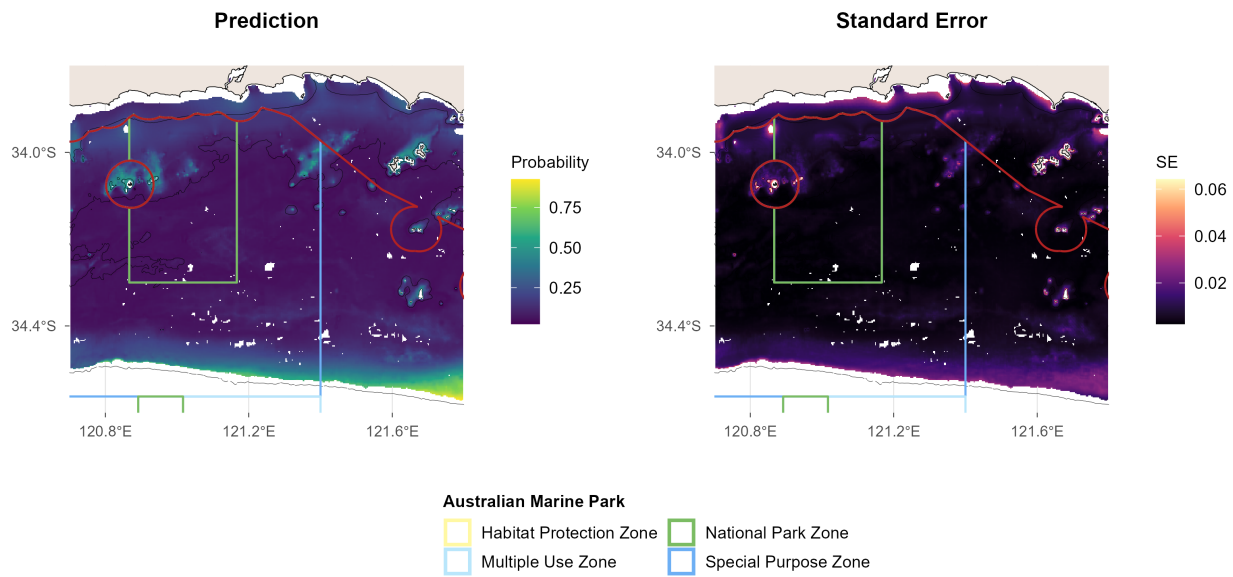


Figure 7: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of demersal fish natural values

Survey effort

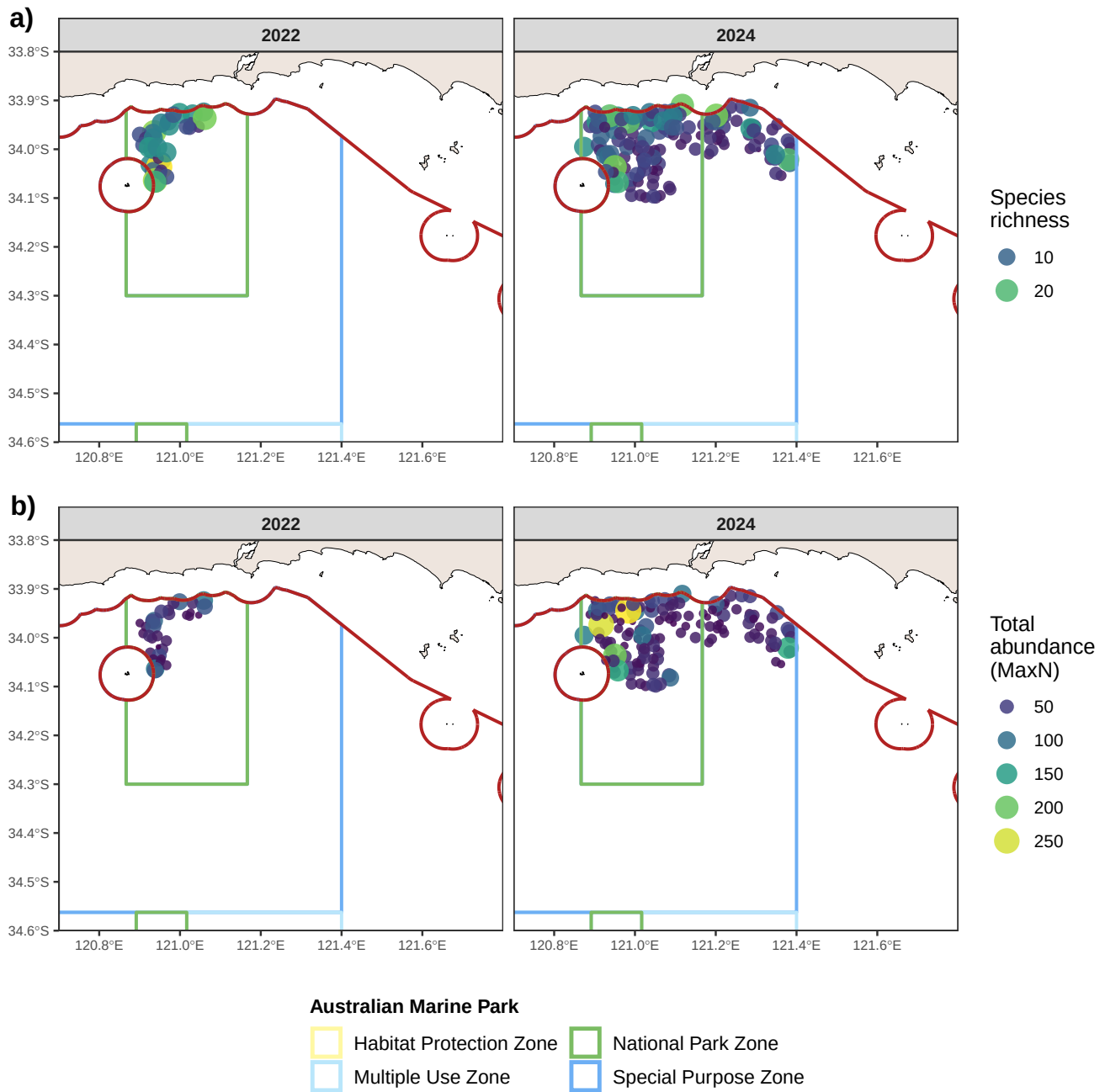


Figure 8: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

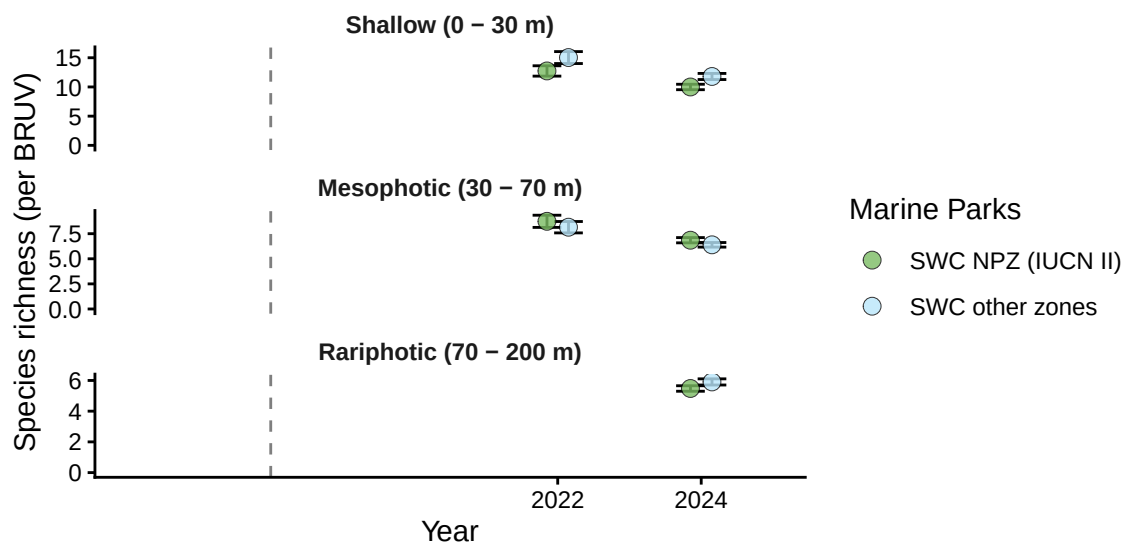


Figure 9.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

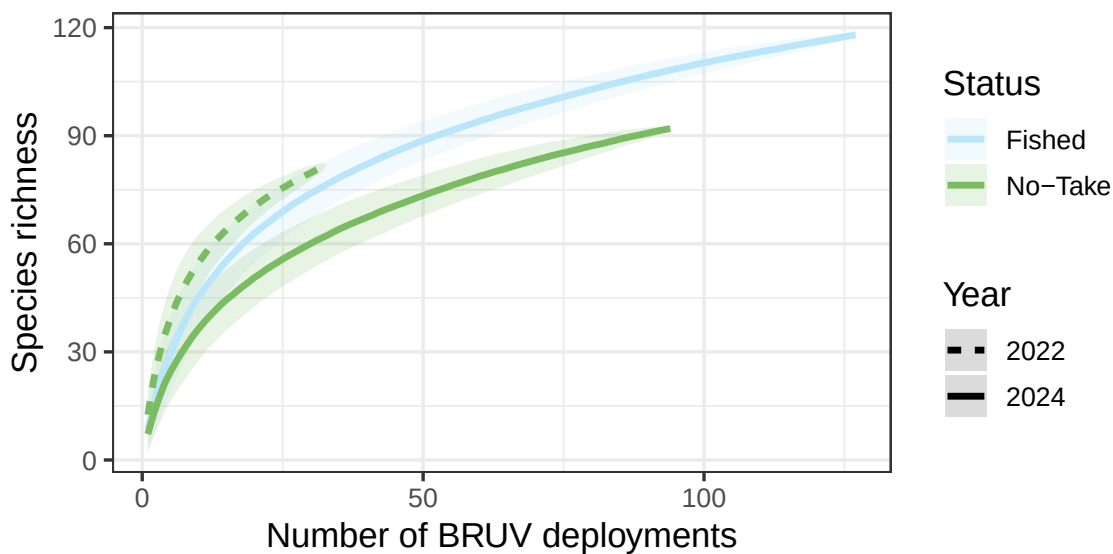


Figure 9.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Species richness

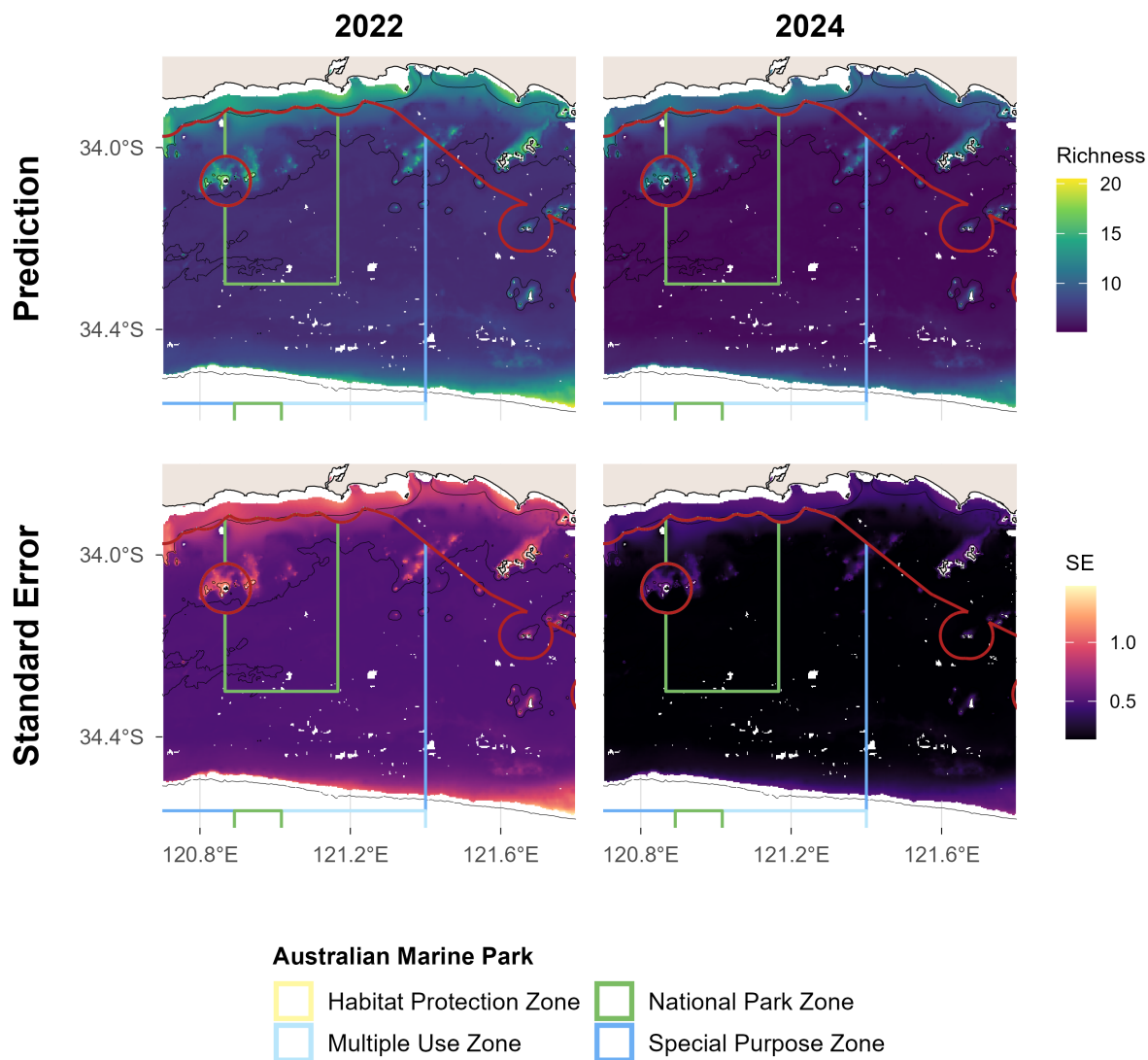


Figure 9.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

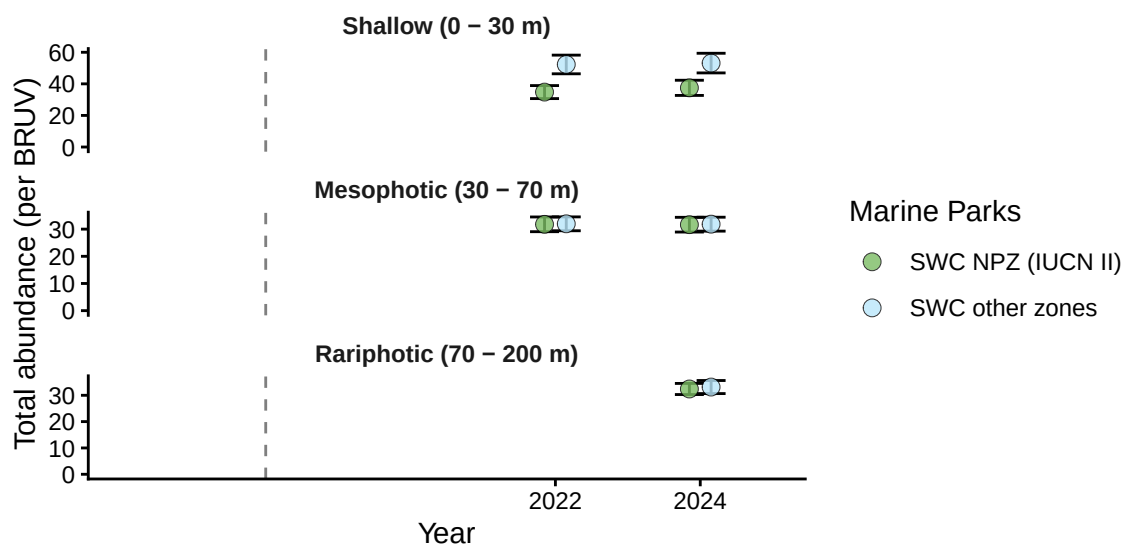


Figure 10.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

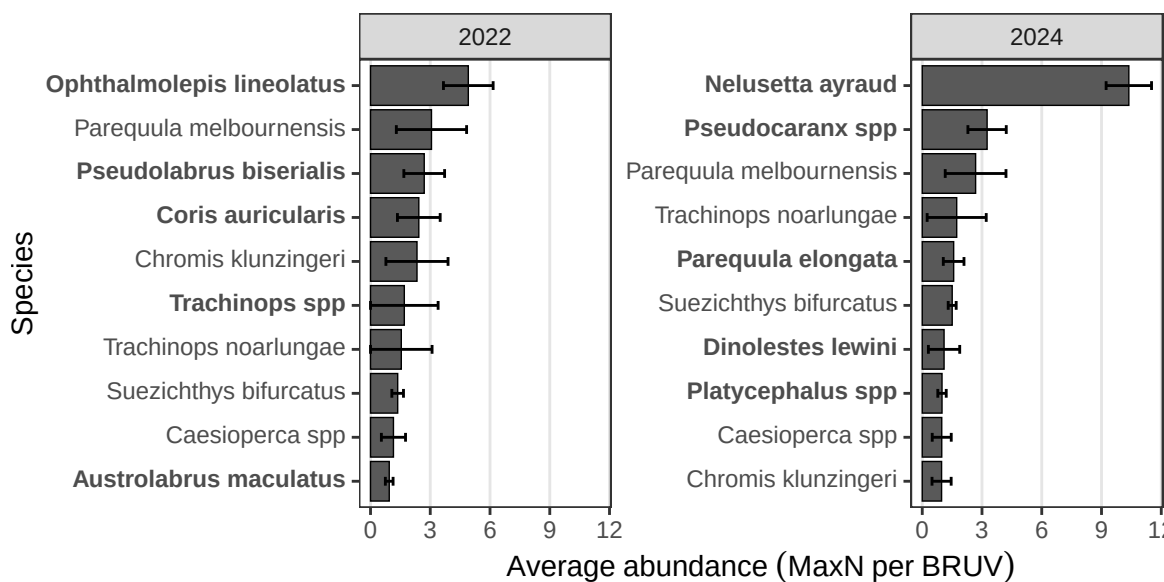


Figure 10.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Total abundance

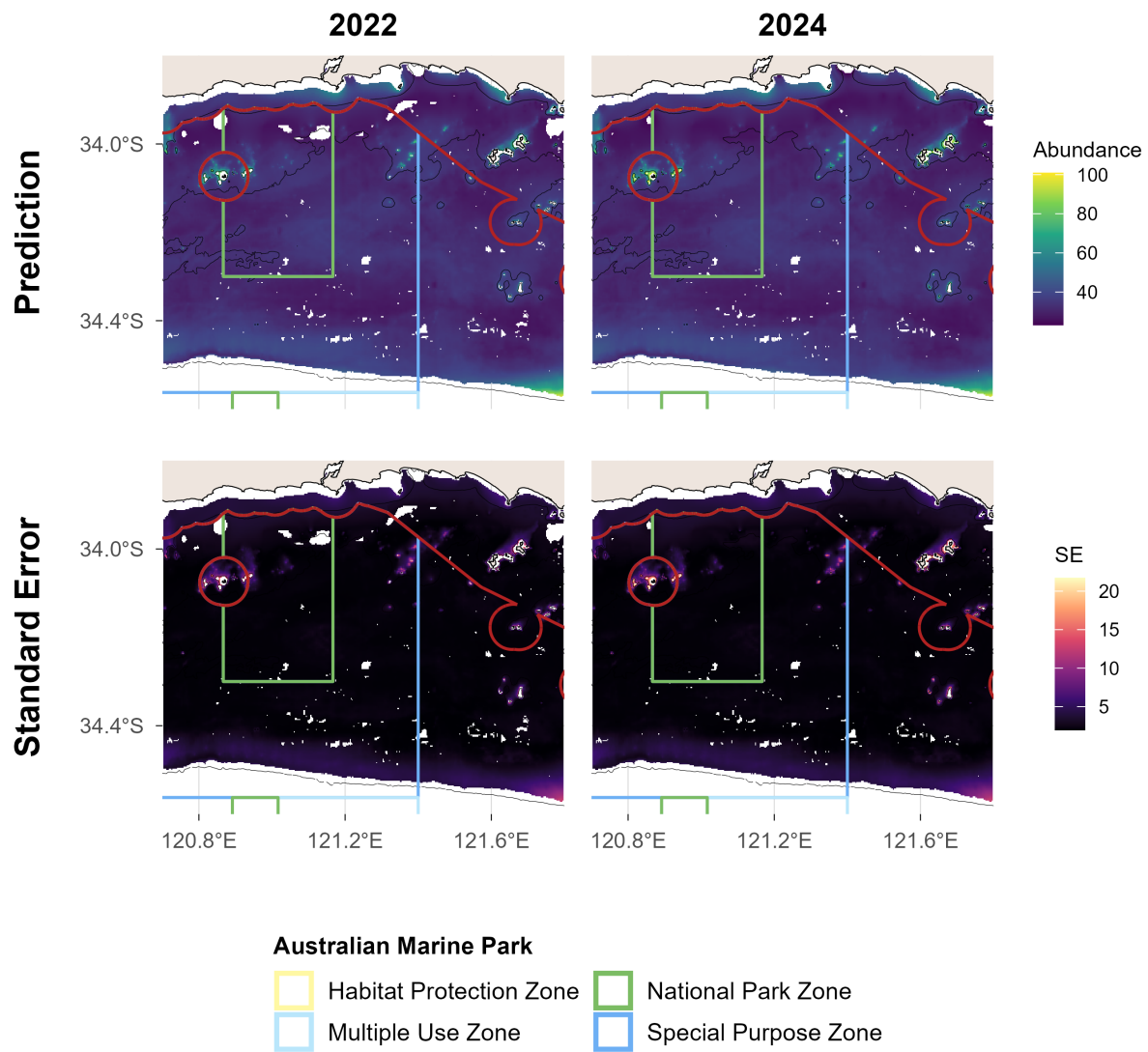


Figure 10.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

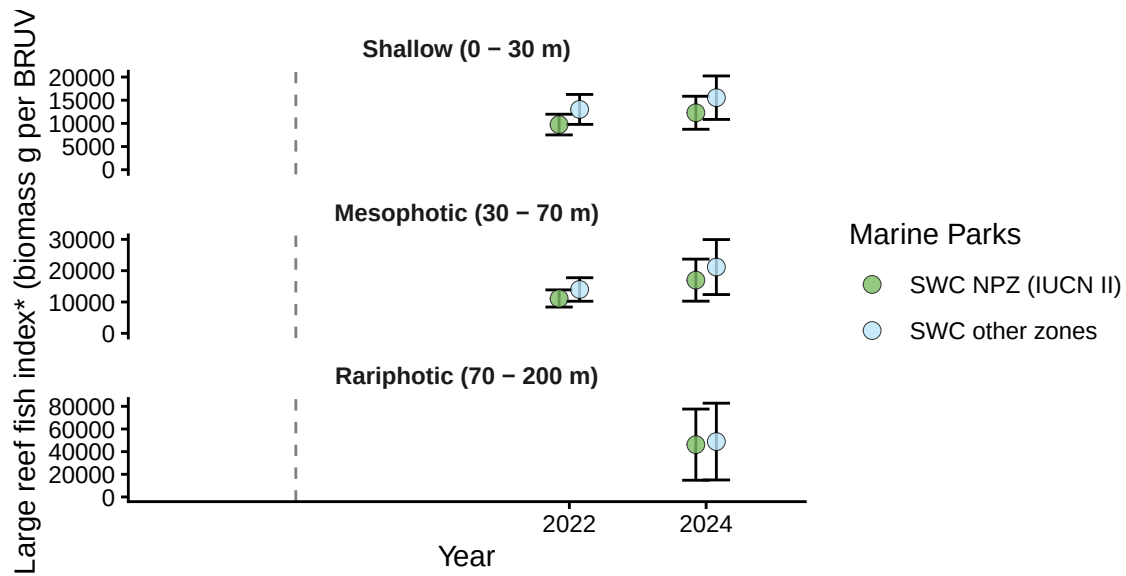


Figure 11.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

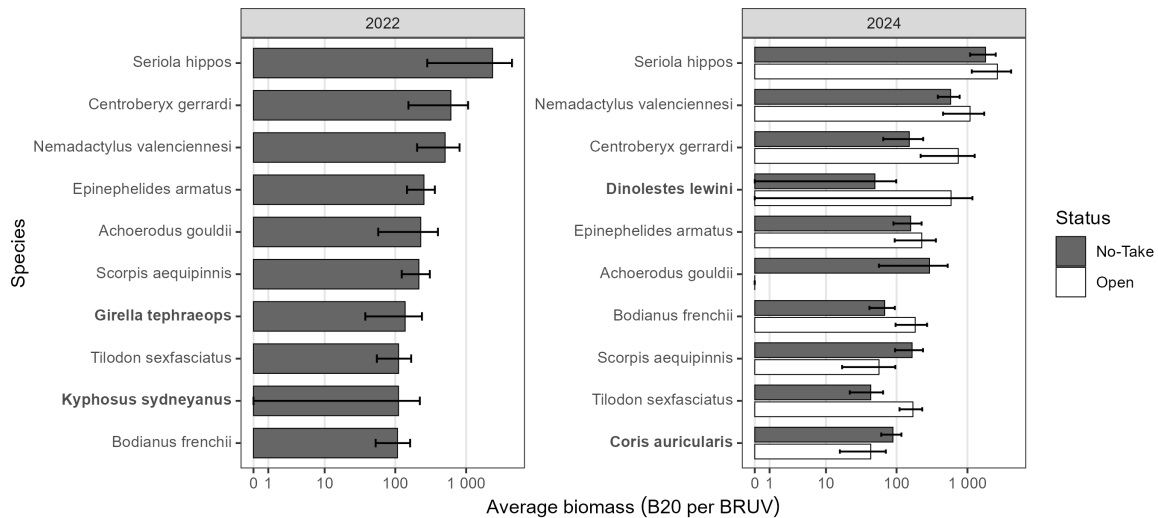


Figure 11.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Large Reef Fish Index*

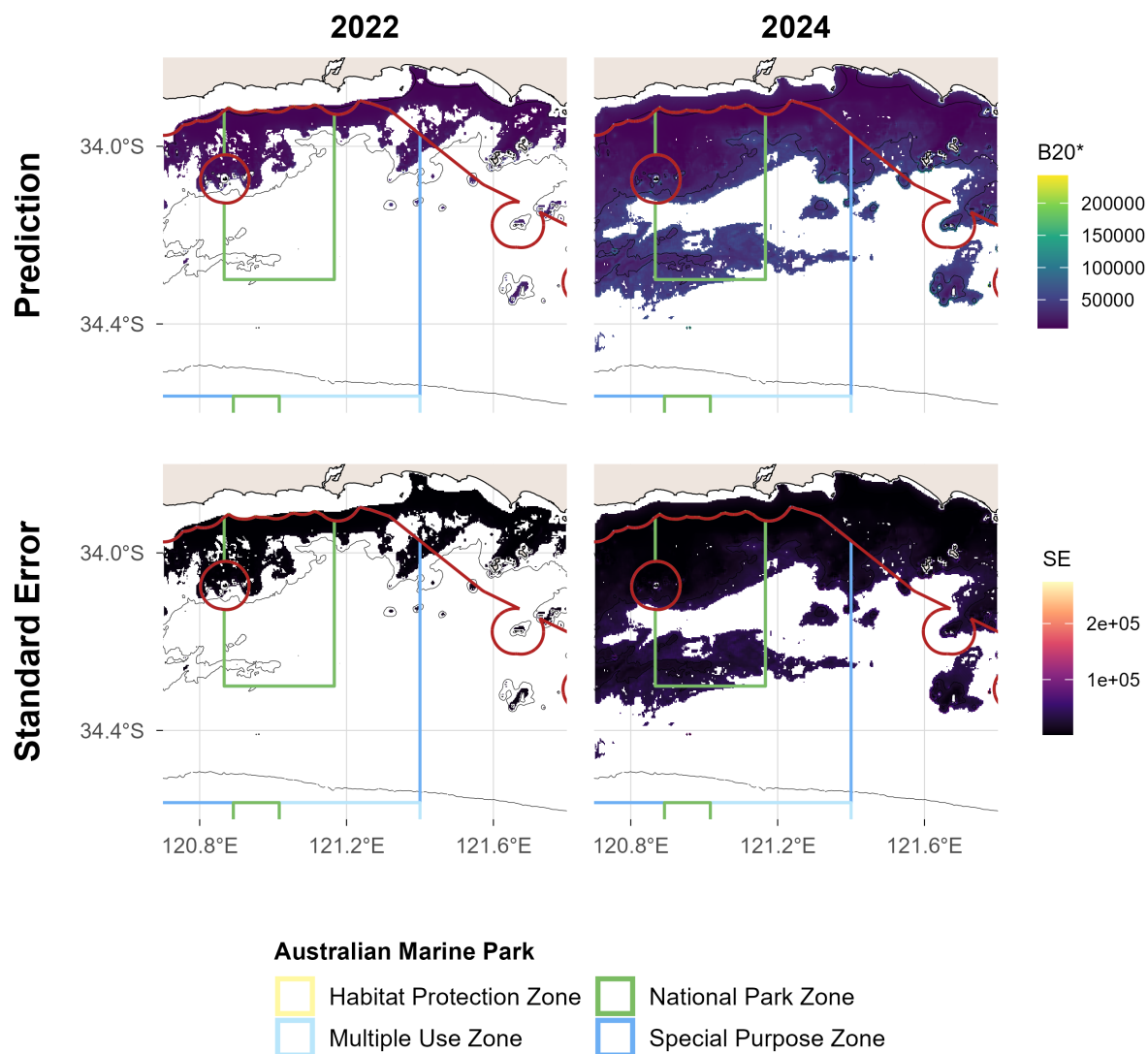


Figure 11.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

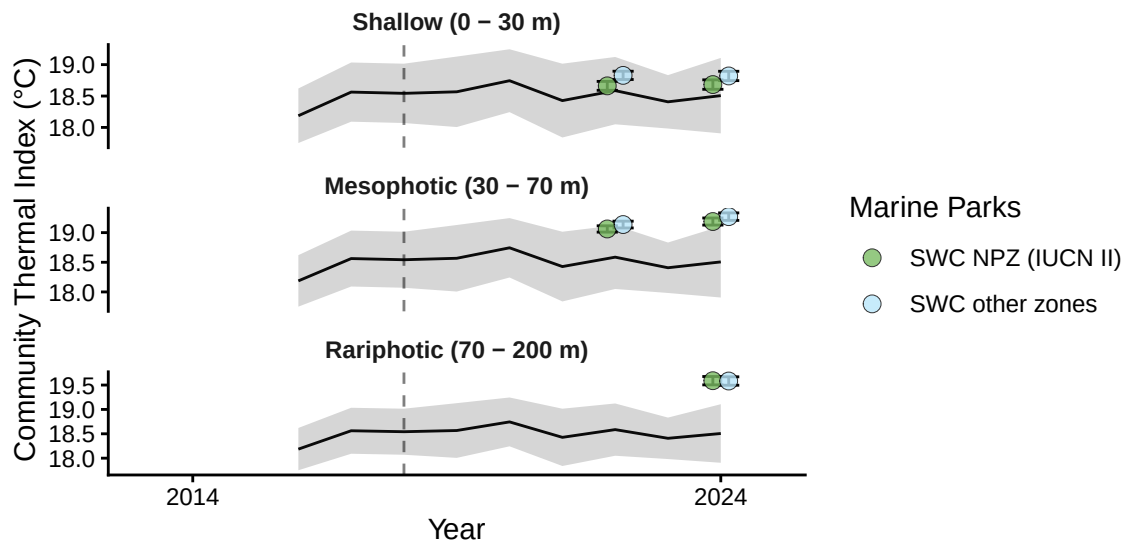


Figure 12.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

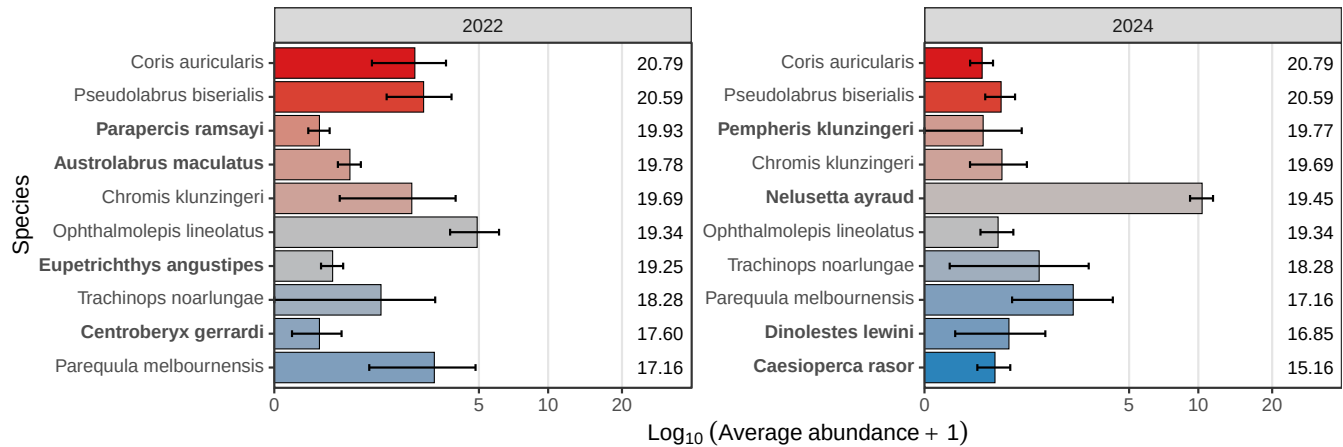


Figure 12.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

Reef fish thermal index

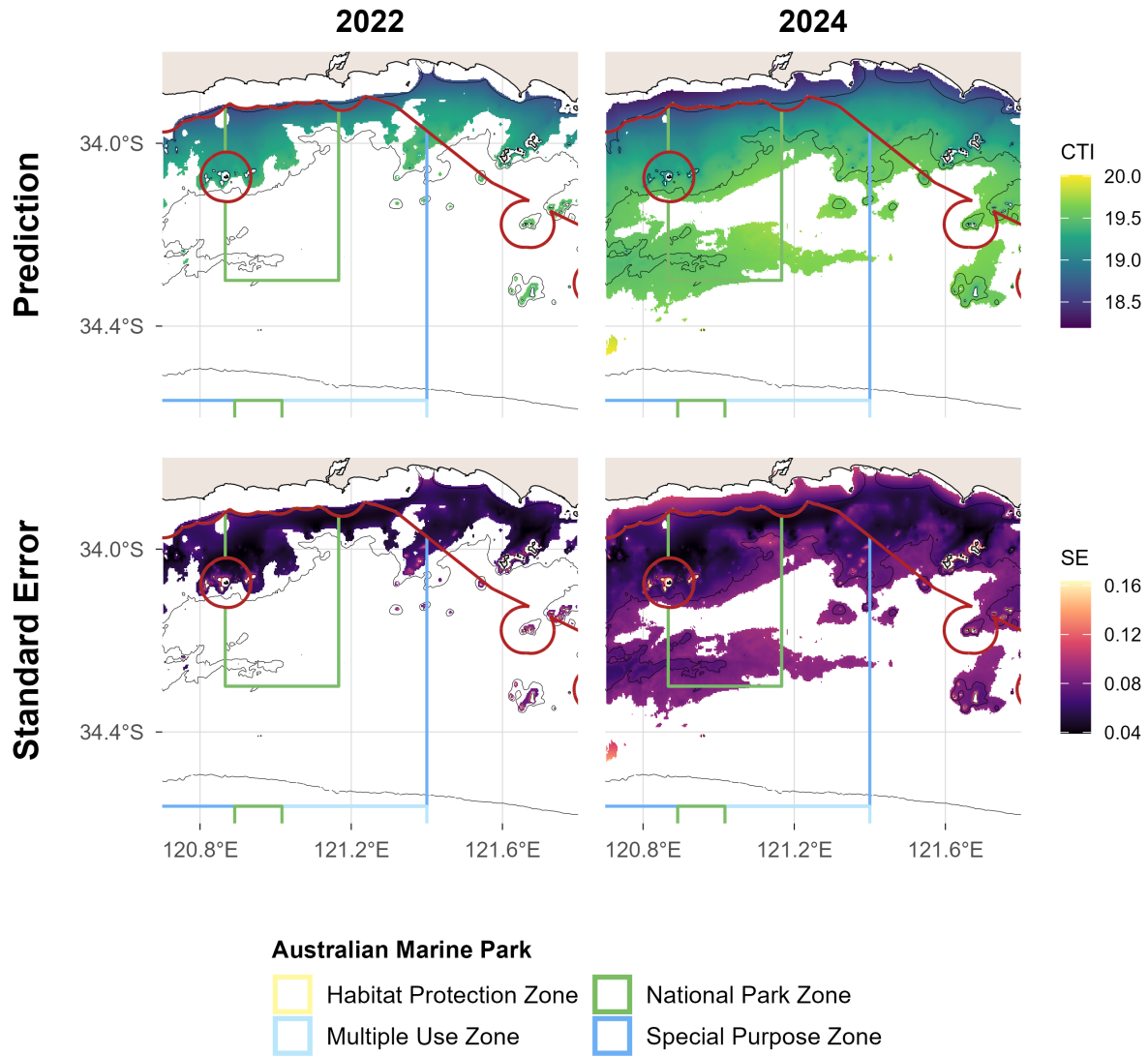


Figure 12.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.